

# Task 1: Business Requirements Analysis

## 1. Company Overview & Organisational Structure

Ubuntu Innovations (Pty) Ltd is a rapidly growing technology startup based in Cape Town, South Africa. The company is relocating to a new office and requires a fully documented IT infrastructure built from scratch across its 25-person workforce.

| Organisational Structure                                                                                                                           | Company Context                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Department</b><br>Executive Management<br>Finance<br>Human Resources<br>Sales & Marketing<br>Software Development<br><b>TOTAL: 25 employees</b> | <b>Industry:</b> Information Technology / Software Development<br><b>Stage:</b> Early-growth startup<br><b>Location:</b> Cape Town, Western Cape, South Africa<br><b>Office:</b> New premises — greenfield IT deployment<br><b>IT Maturity:</b> Low — no existing legacy infrastructure<br><b>Operating Hours:</b> Mon–Fri, 08:00–17:00 SAST<br><b>Users:</b> 25 full-time employees across 5 departments |

## 2. Business Requirements Summary

The following requirements were identified across all departments and must be addressed through the IT infrastructure design, security policies, and support procedures.

| # | Category                     | Requirement                                                           | Priority        |
|---|------------------------------|-----------------------------------------------------------------------|-----------------|
| 1 | <b>Connectivity</b>          | Wired & wireless LAN for all 25 users; business-grade firewall router | <b>Critical</b> |
| 2 | <b>User Management</b>       | Individual accounts with dept-based security groups & RBAC            | <b>Critical</b> |
| 3 | <b>File Storage</b>          | NAS with departmental folder structure and access permissions         | <b>Critical</b> |
| 4 | <b>Printing</b>              | Two shared network laser printers accessible across the LAN           | <b>Medium</b>   |
| 5 | <b>Data Security</b>         | Antivirus, device encryption (laptops), MFA, patch management         | <b>Critical</b> |
| 6 | <b>Backup &amp; Recovery</b> | Daily incremental, weekly full, monthly archive; cloud offsite copy   | <b>High</b>     |
| 7 | <b>Applications</b>          | Microsoft 365 for all staff; dept-specific software per section 3     | <b>Critical</b> |
| 8 | <b>Power Protection</b>      | Two 1500VA UPS units protecting core networking equipment             | <b>High</b>     |
| 9 | <b>Documentation</b>         | Troubleshooting guide and IT support procedures for staff             | <b>Medium</b>   |

## 3. Business Application Requirements by Department

| Department            | Required Software                                         | Notes                                                         |
|-----------------------|-----------------------------------------------------------|---------------------------------------------------------------|
| All Staff (25)        | Microsoft 365, Antivirus (Bitdefender), Zoom, Web Browser | Core productivity and security baseline for every device      |
| Executive Mgmt (3)    | Power BI, Asana, BitLocker encryption                     | Laptops must be fully encrypted; mobility-focused             |
| Finance (4)           | Sage Business Cloud, Advanced Excel, Seclore DRM          | SARS-compliant; payroll data requires strict access control   |
| Human Resources (3)   | SimplePay, SharePoint (M365), DocuSign                    | Employee records need privacy controls and digital signing    |
| Sales & Marketing (7) | HubSpot CRM, Canva Pro, Hootsuite, Google Analytics       | Cloud-dependent; stable internet access is critical           |
| Software Dev (8)      | VS Code, Git, GitHub, Docker, Ubuntu Linux 22.04          | Developer workstations run Linux; elevated permissions needed |

## 4. Operational Challenges

Six key operational challenges have been identified that the IT infrastructure design must address from the outset.

| Challenge                    | Description                                                               | Challenge                    | Description                                                              |
|------------------------------|---------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------|
| <b>Greenfield Deployment</b> | All infrastructure must be built from scratch before operations begin.    | <b>Data Segregation</b>      | Finance and HR hold sensitive data requiring strict access controls.     |
| <b>Load Shedding</b>         | SA power outages risk downtime and data loss without UPS protection.      | <b>Cybersecurity Threats</b> | Phishing and ransomware are high-probability risks requiring MFA and AV. |
| <b>Scalability</b>           | Infrastructure must accommodate projected staff growth without a rebuild. | <b>Software Licensing</b>    | 25 devices require tracked, compliant licences across all applications.  |

## 5. Data Security & Backup Requirements

| Data Security Controls                                                                                                                                                                                                       | Backup & Recovery Schedule |               |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------|----------------|
| Antivirus & endpoint protection on all 25 devices<br>BitLocker encryption on all 5 executive laptops<br>Multi-factor authentication (MFA) for all accounts<br>Security patching within 72hrs (critical) / 14 days (standard) | Frequency                  | Type          | Retention      |
|                                                                                                                                                                                                                              | Daily                      | Incremental   | 7 days rolling |
|                                                                                                                                                                                                                              | Weekly                     | Full backup   | 4 weeks        |
|                                                                                                                                                                                                                              | Monthly                    | Archive       | 12 months      |
|                                                                                                                                                                                                                              | Continuous                 | Cloud offsite | Real-time      |

|                                                                                                                |                                                                                           |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Written Acceptable Use Policy issued to all staff<br>Physical access controls for server room and network rack | <b>RTO: &lt; 4 hours   RPO: &lt; 24 hours</b><br>NAS with RAID-5 as primary backup target |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|

## 6. Connectivity & Network Requirements

| Wired & Wireless Network                                                                                                                                                                                                                                                                                                                 | IP Addressing Plan (Preview) |                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------|
| Gigabit Ethernet for all 25 desktop workstations<br>24-port managed switch for internal LAN<br>2× Wi-Fi 6 dual-band access points for wireless coverage<br>Business-grade firewall router (NAT, VPN, traffic filtering)<br>VLAN segmentation: servers, users, printers, Wi-Fi<br>2× 1500VA UPS units for core equipment power protection | Device Type                  | Subnet          |
|                                                                                                                                                                                                                                                                                                                                          | Servers / NAS                | 192.168.10.0/24 |
|                                                                                                                                                                                                                                                                                                                                          | User Devices                 | 192.168.20.0/24 |
|                                                                                                                                                                                                                                                                                                                                          | Printers                     | 192.168.30.0/24 |
|                                                                                                                                                                                                                                                                                                                                          | Wi-Fi Clients                | 192.168.40.0/24 |

## 7. Key Recommendations

| # | Recommendation                                                                                                                 |
|---|--------------------------------------------------------------------------------------------------------------------------------|
| 1 | Deploy a VLAN-segmented network from day one to ensure security and scalability as the company grows.                          |
| 2 | Implement centralised identity management (Active Directory or equivalent) for user accounts and group policy enforcement.     |
| 3 | Enforce a zero-trust security baseline from the outset: MFA on all accounts, endpoint AV, encrypted laptops, and a signed AUP. |
| 4 | Select all hardware with a minimum 3-year South African vendor warranty to reduce support risk and downtime.                   |
| 5 | Design the backup strategy to satisfy both on-premises (NAS/RAID-5) and offsite (cloud replication) requirements.              |
| 6 | Document all configurations, procedures, and troubleshooting guides immediately to reduce single-point-of-knowledge risk.      |
| 7 | Plan a 30% growth buffer in IP addressing, switch port capacity, and NAS storage to accommodate future hiring.                 |

## Task 2: Hardware Inventory

### Hardware Inventory Table

The table below details all recommended hardware assets for Ubuntu Innovations (Pty) Ltd. All costs are estimated in South African Rand (ZAR).

| Asset                  | Qty | Suggested Specification                                                  | Unit Cost (ZAR) | Total Cost (ZAR) | Business Justification                                 | Warranty                          |
|------------------------|-----|--------------------------------------------------------------------------|-----------------|------------------|--------------------------------------------------------|-----------------------------------|
| Desktop Computers      | 20  | Intel Core i5, 16GB RAM, 512GB SSD, Windows 11 Pro                       | R12,000         | R240,000         | General office use across all 5 departments            | 3-year on-site                    |
| Laptops                | 5   | Intel Core i7, 16GB RAM, 512GB SSD, Windows 11 Pro + BitLocker           | R18,000         | R90,000          | Mobility for Executive Management team                 | 3-year on-site                    |
| Router / Firewall      | 1   | Business-grade firewall router (e.g. Cisco RV340 / Fortinet 40F)         | R8,500          | R8,500           | Secure internet access, NAT, VPN and traffic filtering | 3-year vendor support             |
| Managed Switch         | 1   | 24-port Gigabit managed switch (e.g. Cisco SG350-28)                     | R4,500          | R4,500           | Internal LAN for all 25 wired devices plus spare ports | 3-year vendor support             |
| Wireless Access Points | 2   | Dual-band Wi-Fi 6 AP (e.g. Ubiquiti UniFi U6-LR)                         | R3,200          | R6,400           | Full wireless coverage across the office floor         | 2-year manufacturer               |
| Network Printers       | 2   | Monochrome laser printer with network card (e.g. HP LaserJet Pro M404dn) | R5,500          | R11,000          | Shared departmental printing — low cost per page       | 1-year; service contract optional |
| NAS Device             | 1   | 8TB RAID-5 NAS (e.g. Synology DS423+ with 4x4TB WD Red drives)           | R14,000         | R14,000          | Centralised file storage and daily backup target       | 3-year manufacturer               |

|                                       |   |                                                     |        |                 |                                                                    |                        |
|---------------------------------------|---|-----------------------------------------------------|--------|-----------------|--------------------------------------------------------------------|------------------------|
| <b>UPS Units</b>                      | 2 | 1500VA UPS<br>(e.g. APC<br>Smart-UPS<br>1500VA LCD) | R3,800 | R7,600          | Power<br>protection for<br>network rack<br>during load<br>shedding | 2-year<br>manufacturer |
| <b>TOTAL<br/>ESTIMATED<br/>BUDGET</b> |   |                                                     |        | <b>R382,000</b> |                                                                    |                        |